

THE UNIVERSITY OF CHICAGO

IMPACT OF MENTAL REPRESENTATION ON CONSUMER BEHAVIORS:
IMPLICATIONS FOR MENTAL BUDGETING AND PREDICTION ALGORITHM
PREFERENCES

A DISSERTATION SUBMITTED TO
THE FACULTY OF THE UNIVERSITY OF CHICAGO
BOOTH SCHOOL OF BUSINESS
IN CANDIDACY FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY

BY
LIN FEI

CHICAGO, ILLINOIS

JUNE 2023

Table of Contents

List of Tables	iii
List of Figures	iv
Acknowledgements	v
Overview	1
Introduction	2
Chapter 1: Consumers' Mental Representation of Expenditures: Implications for Spending and Savings Decisions	5
Abstract	6
Study 1: Representation of Expenditures	19
Study 2: Taxonomic Distance and Spending Adjustment	25
Study 3: Taxonomic Distance and Promotions	36
Study 4: Examination of Grocery Purchase	42
Chapter 2: Prediction by Replication: People Prefer Prediction Algorithms That Replicate the Event Being Predicted	54
Abstract	55
Study 1: Prediction of Die Roll Outcomes	65
Study 2: Prediction in Real-world Domains	75
Study 3: Multiple Predictions and Choices	83
Study 4: Simulation as Intervention	88
General Discussion	90
Appendix 1: Tables	96
Appendix 2: Figures	103
References	116
Supplemental Files: Web Appendix	129

List of Tables

Table 1.1: Stimuli from Study 2b, Chapter 1	96
Table 1.2: Stimuli from Study 3a and 3b, Chapter 1	97
Table 1.3: Product Sets from Study 4, Chapter 1	98
Table 2.1: Recommendation Description from Study 2c, Chapter 2	99
Table 2.2: Dimension Wording from Study 2c, Chapter 2	100
Table 2.3: Conditions from Study 3, Chapter 2	101
Table 2.4: Regression from Study 3, Chapter 2	102

PREVIEW

List of Figures

Figure 1.1: Possible Representation of Expenditures, Chapter 1	103
Figure 1.2: Successive Pile Sort Interface, Chapter 1	104
Figure 1.3: Multidimensional Scaling Reduction with Clustered Groups from Study 1, Chapter 1	106
Figure 1.4: Dendrogram of the Products on Aggregate Level from Study 1, Chapter 1	107
Figure 1.5: Spending Adjustment on Comparison Items from Study 2a, Chapter 1	108
Figure 1.6: Spending Adjustment on Comparison Items from Study 2b, Chapter 1	109
Figure 1.7: Raw Ratings and Estimated Marginal Means of Study 2c, Chapter 1	110
Figure 1.8: Proportions Choosing the Closest Comparison Product from Study 3, Chapter 1...	111
Figure 1.9: Regression Coefficients by Year on Close Focal vs Far Focal from Study 4, Chapter 1.....	112
Figure 2.1: Likelihood-to-Use and Replicativeness Rating from Study 1b, Chapter 2	113
Figure 2.2: Replicativeness & Willingness-to-Use Ratings from Study 2c, Chapter 2	114
Figure 2.3: Percentage Choosing the Target Algorithm from Study 3, Chapter 2	115

Acknowledgements

I am extremely grateful to my co-advisors Dan Bartels and Berkeley Dietvorst, for showing me how rigorous and scientific research is and should be, for challenging me to articulate and explain my thinking, and for always giving thoughtful advice and words of encouragement. I am also incredibly thankful to Luxi Shen, whose mentorship and friendship have been essential to my research experience and personal development, as well as Reid Hastie, who has always been so generous with his time and wisdom and has always given out careful thoughts to my projects. I deeply appreciate my fellow cohort Danny Katz and Minkwang Jang who are always there for their peer, as well as the HOCAG group which is always an encouraging and fun environment to discuss research.

Finally, I am extremely grateful for Ignatius Liu, who has been incredibly supportive of me as a researcher and a person. Your help and your company have contributed much to this dissertation and this wonderful journey.

Overview

People constantly encode and represent the information around them, and they make various inferences, judgments, and decisions based on their representation. The dissertation aims to connect mental representations to consumer behaviors. Investigating how consumers represent their choices and contexts allows us to generate new predictions about their behaviors and provide novel insights into the cognitive processes under these enriched contexts. In Chapter 1, I explore the relationship between the representation of concepts and spending behaviors. In Chapter 2, I explore how people represent prediction events and consequently how it influences their prediction algorithm choices. Additional norming studies and detailed study stimuli are available in Supplemental Files: Web Appendix.

Introduction

People constantly encode and represent the information around them, and they make various inferences, judgement, and decisions based on their representation. For example, people could use groups of exemplars or prototypes to represent concepts, could leverage maps or networks to represent relationships, and could adopt mental models or scripts to represent processes. The study of mental representations is foundational in understanding the human mind because it is central to how humans process and make use of information.

Understanding how people represent the world around them has much merit particularly in consumer contexts (Bartels and Johnson 2015). A coherent theory on people's representation can help highlight the potential common ground of seemingly distinct findings and unify multiple established phenomena. For example, how people represent monetary values can provide an explanation to concave utility function, loss aversion, and hyperbolic discounting (Stewart, Charter, and Brown 2006). In addition, understanding consumer's representations allows us to further predict their reactions and behaviors in contexts like new product launches (Moreau, Markman, and Lehmann 2001), product bundles, and service failures. In applying representation to the study of consumer behaviors, we simultaneously learn about the behavior and the thinking behind it.

More generally, identifying the role of representation and cognition complements the common economic approach to studying consumer behavior. The study of consumer behaviors has long been focused on how people's behaviors and judgments deviate from those that are economically optimal (Thaler 2015). The cognitive approach on the other hand puts much emphasis on why people are behaving the way they are and where the phenomena come from. Combining the two approaches provides a better understanding of why consumers make

decisions in particular ways and generates predictions on when consumers will deviate from the economic prediction.

Studying representation with these enriched environments and complex decision problems in consumer contexts also provides new insights into the human mind. Consumers face some of the most complicated decisions such as purchasing a house, and the study of these choices invites new theories on problem solving and decision making (Payne, Bettman, and Johnson 1993). For example, the study of brand and product categorization expands our understanding on how a category prompts the objects in it (Nedungadi 1990), and the study of context effects in product choices (Huber, Payne, and Puto 1982) has inspired much theorization on how contexts shape decisions (Trueblood et al. 2014).

Finally, connecting representation with behavior allows us to identify contexts where behaviors pertaining to daily lives, spending, and decisions, and thinking are related. In this dissertation, I aim to identify two of these contexts. Specifically, I found that representations influence people's behaviors in how they manage their budgets and how they choose prediction algorithms, in ways that are distinct from predictions of simple normative models.

In Chapter 1, I explore the relationship between the representation of concepts and spending behaviors. Through 4 sets of studies, I find that when consumers deviate from budget on one item, they are more likely to adjust spending for items closer in representation than further. In Study 1, I recover the taxonomy that people use to represent common expenditures with a successive pile-sort paradigm. Study 2 found that recovered taxonomy predicts people's self-reported spending adjustment, while Study 3 and 4 found that the taxonomy predicts people's consequential choices and actual grocery purchase behavior. This chapter provides evidence that representation influences spendings, even when money is theoretically fungible.

In Chapter 2, I explore how people represent prediction events and consequently how they choose prediction algorithms. Through 4 sets of studies, I found that people prefer to use prediction algorithms that replicate the same process of the even being predicted, which implies that the mental models of a prediction event include the exact process through which the outcome is generated. I found that when predicting both simple but random tasks (Study 1) and real-world outcomes (Study 2), people prefer algorithms that replicate the process of the prediction event, even when the algorithms might prefer worse. Study 3 provides a boundary of this preference, while Study 4 designs and tests an intervention based on this preference. This chapter highlights how people's representation changes consumers' preferences and choices.

**Chapter 1: Consumers' Mental Representation of Expenditures: Implications for Spending
and Savings Decisions**

PREVIEW

Abstract

People's mental representation of expenditures is crucial to how they budget. We propose that much like how people represent natural objects, people represent expenditures in a hierarchical taxonomy. Across seven studies, we found evidence of a hierarchical representation of expenditures. We first recover people's mental representation using a successive pile-sort method that asks people to form hierarchies of categories with common expenditures (e.g., rent, dining out, etc.). We found that there is consensus in people's hierarchical representations of expenditures and that their representations are relatively stable over time. Further, we found that people's adjustment in their spending behavior can be predicted by the distance between items in their representation. Specifically, when people overspent on an item, they were more likely to spontaneously adjust spending for items closer in representation than further. We examine this spontaneous adjustment behavior using both lab studies and field data with over 7-million grocery shopping trips. The findings highlight the connection between mental representation and consumer behavior, and emphasize the importance of studying cognitive representation in the context of consumption.

Consumers often maintain budgets. That is, they set restrictions on how much they wish to spend, and they behave as though money is nonfungible (Thaler 1985; 1999) and earmarked for specific functions (e.g., gas, food; Antonides et al. 2011; Hasting and Shapiro 2013, 2018; Heath and Soll 1996; Cheema and Soman 2006; Soman and Cheema 2011; Thaler 1999). For example, consumers might set a budget specifically intended for gas. When gas prices drops, consumers, who now have a gas-budget surplus, may use it to purchase premium gas (Hastings and Shapiro 2013). In a recent survey, 80% of people reported that they have formally or informally budgeted at some point in their lives, and the practice of budgeting is positively correlated with financial wellbeing (Zhang et al. 2020). Therefore, understanding how budgeting works is essential for understanding consumer financial decision making and consumer welfare.

Research on budgeting mostly theorizes that consumers group expenditures into a single level of budgeting categories (Heath and Soll 1996; Cheema and Soman 2006, Zhang et al. 2020, cf. Henderson and Peterson 1992). Some consumers might group items into relatively general categories such as “Food” and “Entertainment”, while others group to more detailed levels such as “Groceries”, “Dining Out”, and “Movies” (Zhang et al. 2020). In this paper, we propose and test a more complete account that people mentally represent expenditures in hierarchical taxonomies. In other words, people represent expenditures in multiple, nested levels of categories where lower levels of the categories are more specific and higher levels are more general. For example, people might think of cereal as “breakfast food” before they think of it more generally as “food” (Markman, Brendl, and Kim 2007), and hence the “breakfast food” category is nested within the higher-level “food” category (Figure 1).

Consequently, we hypothesize that the taxonomic distance between items is important for people’s spending decisions in situations where they deviate from their budget. We define

taxonomic distance as the level at which the expenditures are categorized together in consumers' taxonomies, and we predict that when people's spending on an item deviates from their budget, they will adjust more on the items that are taxonomically closer than further. This prediction is important to marketers who are interested in understanding the type of products consumers adjust and the degree of adjustment when consumers encounter spending deviations such as holiday splurges. Further, past research that hypothesizes single level budgeting categories cannot account for this adjustment. In this paper, we adopt a method from cognitive anthropology to approximate the represented taxonomy of expenditures. Then, we use the taxonomy to predict how people adjust their spending in various contexts, including real shopping behaviors.

Theoretical Background

Connecting Mental Representation to Mental Accounting

Mental accounting is grounded in the way that people categorize and represent concepts (Henderson and Peterson 1992). Mental accounting refers to the cognitive operations by which prospects are evaluated with respect to some specific "topical" account instead of a "comprehensive" account (Linville and Fischer 1991; Thaler and Johnson 1990; Kahneman and Tversky 1984; Soman 2004; Thaler 1999; Zhang et al. 2020). For example, people are more likely to travel to save \$5 on a calculator that originally costs \$15 than when it costs \$125. This is likely because they evaluate the \$5 with respect to the total "topical" expenditure (\$15 or \$125) but not their total wealth. In doing so, people are associating money with different expenditure concepts, and how people represent the concepts will consequently influence how they spend the resources.

Researchers have connected mental representation with mental accounting (Arkes et al. 1994; Levav and McGraw 2009; Evers, Imas, and Kang 2022) and have provided insights on how people form and use mental accounts. For example, representation of the source of money influences the types of mental accounts that are constructed (Levav and McGraw 2009, Reinholtz, Bartels, and Parker 2015): money associated with negative emotions is less likely to be spent on hedonic purchases, and money associated with brand-specific gift cards is more likely to be spent on brand-typical items. Representation of expenditures also influences the membership of expenditures with respect to a mental account. Expenditures that have a high degree of feature-based (i.e., attribute) similarity are often integrated together as a larger loss (Evers, Imas, and Kang 2022).

Similarly, mental representation has provided much insight into how people mentally budget (Henderson and Peterson 1992; Heath and Soll 1996). If people mentally represent expenditures, for example, they would spontaneously categorize expenditures and organize them in categories when prompted to think about budgets, which is what researchers have found (Cheema and Soman 2006; Sussman and Alter 2012). The expenditures in a category are often similar to each other in some respects: they may share similar features (e.g., electronics) or fulfill similar goals (e.g., entertainment; Felcher, Malaviya, and McGill 2001). Expenditures that are grouped into a budgeting category have graded membership—the typicality of an expenditure within a category can differ (Reinholtz et al. 2015), which is another prediction from research on mental representation. For example, shirts can be more typical of the clothing category than gloves (Heath and Soll 1996).

However, previous research has not connected the hierarchical nature of representation with mental accounting. To the authors' knowledge, only one paper has provided preliminary

tests that people represent mental accounts hierarchically (Henderson and Peterson 1992). Specifically, Henderson and Peterson (1992) mentioned that people have a larger budget for higher-level categories but did not provide predictions on how people make spending decisions when there are hierarchical budgets. However, the researchers did not make concrete predictions on how the hierarchy influences budgeting behavior. In the sections below, we detail the reasoning for proposing a hierarchical representation of expenditures, and we highlight additional contributions of introducing a hierarchical representation in budgeting.

Hierarchical Representation of Expenditures

There are several reasons why we think consumers group expenditures hierarchically. First, research suggests that people represent concepts of the natural world and of their everyday lives in taxonomies (Berlin 1992). People may group cats and dogs into the category “mammals” and then group mammals together with birds and fish into the superordinate category “animal”. Organizing concepts at multiple, nested levels is a cognitively efficient way for people to store and access information related to different objects (Collins and Quillian 1969; Smith 1978) and generalize the information to more specific concepts (e.g., a dog inherits the properties of a mammal; Osherson et al. 1990). Research suggests that people hierarchically represent animals (e.g., Lopez et al. 1997), trees (Collins and Quillian 1969; Medin et al. 1997), furniture (Rosch et al. 1976), and clothing (Rosch and Lloyd. 1978), and we suspect that people represent concepts of expenditures in terms of multiple, nested categories as well.

Further, people also represent objects in a way that reflects their experience and their knowledge of the world (Markman 1999; Medin and Atran 2004). For example, the taxonomy for trees that landscapers use tends to reflect differences between trees in shape and appearance, while botanists focus more on the scientific taxonomy. Each group’s taxonomy reflects the

knowledge of trees that is most useful for the way that each group typically interacts with and thinks about trees. Similarly, people's interactions with consumer goods likely also shape the categories they use to represent them. For example, grocery items are often subdivided into aisles that serve different functions (breakfast food, frozen meals, paper products, etc.). Further, aisles typically have different sections dedicated to smaller groupings of products (e.g., baking goods aisle may contain both kitchen appliances and flour). As consumers navigate through these shopping environments, they will likely internalize this organizational scheme into their representation.

Finally, previous research on consumer behavior has hinted that people represent expenditures hierarchically. Several studies have suggested that consumers represent snack foods and fast-food brands in taxonomies (Ratneshwar and Shocker 1991; Nedungadi, 1990). Specifically, Ratneshwar and Shocker (1991) recovered a hierarchical structure from their similarity rating data. Nedungadi (1990) investigated the relationship between brand hierarchy and brand recall. In this paper, we extend the view that people represent objects hierarchically and apply it to spending and budgeting.

Some Contributions of a Taxonomy-Based Theory of Mental Budgeting

The notion that expenditures are represented in terms of a taxonomy addresses some limitations in the existing literature on mental budgeting. First, the current budgeting literature often assumes that people have predefined budgeting categories. Papers that investigate the type of budgets people construct often elicit those categories in a top-down fashion, asking directly for the budgeting categories people have in mind (Heath and Soll 1996, Zhang et al. 2020). However, people might come up with the most common categories they budget for (e.g., "food"), and hence the listed budgets might often miss categories like exceptional expenditures

(Sussman and Alter 2012). Relatedly, previous papers on mental budgeting also often prompted people with pre-defined categorical labels when investigating budgeting behaviors (Heath and Soll 1996; Cheema and Soman 2006). For example, participants would be examining their budgets for “food” and for “entertainment”, which are researcher-defined categories, when they are deciding how to categorize their expenditure of “dining out”. Yet not every constructs these specific budgeting categories (Zhang et al. 2020), and we have limited insights into how people budget in absence of these given categories.

In addition, much of the budgeting literature suggests that consumers have defined expenditure membership with respect to the predefined budgeting categories. Consumers tend to adjust more on items that are typical to these categories (Heath and Soll 1996, Reinholtz et al. 2015), and are worse at evaluating expenditures that are exceptional (Sussman and Alter 2012). However, it is still unclear how consumers define the typicality or exceptionality of an expenditure. Further, consumers often are flexible in the way they categorize expenses (Cheema and Soman 2006) and might categorize a specific expenditure differently depending on the context (“dining out” can belong to “food” or “entertainment”). This suggests that consumers might not have clear categorical membership for expenditures.

A hierarchical representation, on the other hand, puts less emphasis on how expenditures relate to categories and focuses more on how expenditures relate to each other. In this paper, we recover the representation through a bottom-up approach that asks people to group expenditures into a hierarchy. We therefore only argue that the distance in the hierarchy matters for people’s budget and do not need to make assumptions on the categories consumers construct or the membership of expenditures. Further, distances in a hierarchical representation can also reveal expenditures that are clustered together. This complements the current literature by providing

insight into the expenditures that are often thought of together and the categories that people naturally construct. With a hierarchical representation, the recovered structure will also reveal the items that are exceptional (e.g., items that are higher up in the hierarchy) or items that can be flexibly categorized (e.g., items that are in the middle of different concepts), which provides additional insight into people's budgeting behaviors.

The ideas in the current manuscript add new, refined predictions about how people adjust their spending when they have overspent or underspent on an item. We first extend the framework of budgeting beyond binary categorical restrictions. Heath and Soll (1996) theorized that consumers set budget categories, and then track and post spending with respect to the category. In this case, spending decisions are binary: when consumers overspent with respect to their budget for a category, they were more likely to adjust their consequent spending on other items in the category than those outside of the category. However, people's purchase behaviors could be influenced by their out-of-category purchases (Chintagunta and Halder 1998; Manchanda et al. 1999). For example, the purchases of soup and yogurt, which are presumably of different product categories, influence each other (Chintagunta and Halder 1998). Therefore, the framework of budgeting should extend beyond binary categorical restrictions. Our proposed hierarchical representation allows us to examine expenditures at different taxonomic distances beyond binary categorical restrictions. We make a concrete prediction that when a consumer deviates from their budget on a spending, they will adjust more on items that are closer in taxonomic distance than further.

More generally, our paper aims to expand on the theorization of mental accounting beyond the binary distinction of "topical accounts" and "comprehensive accounts" (Kahneman and Tversky 1984; Thaler 1999). Mental accounting is often seen as the behavior that "relates the

consequences of possible choices to a reference level that is determined by the context within which the decision arises” (Thaler 1999). For example, people evaluate a discount of a calculator against the topical account of a calculator purchase rather than against their comprehensive wealth. In contrast, a hierarchical structure implies that there could be multiple nested reference accounts, and people might recruit different reference levels depending on the expenditures they are evaluating. In other words, consumers might be referencing nested mental accounts on a more continuous spectrum than just “topical” and “comprehensive” accounts.

Our prediction not only extends the budgeting literature but also provides insight into consumer research related to cross-product elasticities. Specifically, we predict that after spending on some items, people might adjust down on taxonomically close items due to their budget restraint, resulting in a pattern that is similar to positive cross elasticity. Much examination on cross-product elasticities either recovers the relationship from an observed dataset (Tian, Lautz, Wallis and Lambiotte 2021), or theorizes that the cross-product elasticities were results of substitutes and complements (Koszegi and Matejka 2020). Our theorization suggests an alternative mechanism that cross-product elasticities could emerge from the influence of taxonomic distance on budgeting behaviors. The construct of taxonomic distance is relevant but distinct from substitutability, because it requires people to evaluate their spending with respect to a budget. Further, a close association between items (i.e., a smaller taxonomic distance) might form because of multiple factors. Items closely associated might overlap in their usage contexts (Ratneshwar and Shocker 1991) or related to the same spending goals (Barsalou 1983, 1985). Therefore, people might group items together because the items are substitutes

(e.g., burger and pizza), complements (e.g., shirt and jeans), both (e.g., detergents and soap) or neither (e.g., sunglasses and watch)¹.

We note that the theory of a hierarchical representation in budgeting is an expansion but not an alternative account to the current budgeting literature. We replicate and extend the finding that consumers construct budgeting categories and within a predefined category, expenditure characteristics like typicality matter for spending behaviors. Our paper tests whether consumers also naturally construct categories in a nested fashion, and such hierarchical representation allows us to examine and address open questions within the budgeting literature.

Implications of a Hierarchical Taxonomy

Although we cannot directly examine how people represent expenditures in their minds, we test several implications from our proposal that people represent expenditures hierarchically. First, we can test if consumers have a shared understanding that some items are more closely related than others. For example, we could evaluate whether consumers share the understanding that shampoo and sunscreen are grouped together at a more specific, lower level than shampoo and movie tickets. We test this commonality between people's understanding by testing whether there is statistical consensus among their representations. The term "consensus" is adopted from the Cultural Consensus Model (CCM; Romney, Weller, and Batchelder 1986, Medin et al. 1997) to suggest statistical agreement or commonality across individuals' representations. The CCM establishes consensus by testing whether participants' representations are positively correlated, and whether the first latent root of the interpersonal correlation matrix is large (Weller, 2007). A consensus between participants would suggest not only that people can represent expenditures in

¹ These examples are from our Study 2c, where we collected substitutability and complementarity ratings for all the product pairs that share the smallest taxonomic distance. The details of the collection method are reported in Study 2c.

a taxonomy, but also that this representation is relatively natural to majority of people. Past research has found that members of a culture have a consensual understanding of animal categories (Lopez et al. 1997) and consumers largely agree on product hierarchies of snack foods (Ratneshwar and Shocker 1991).

However, it is also possible that people have systematically different representations and that there might not be consensus across consumers. Literature has found that representations of the same objects can vary across cultures (Lopez et al. 1997) and occupations (Medin et al. 1997). For example, the way people represent animals and trees is influenced by their environment and the functions the objects fulfill. If there are large differences between consumer segments, there could be discrepancy in how people represent expenditures. Therefore, we first explore the competing hypotheses of whether there is consensus in people's representations.

H1a: Consumers represent expenditures similarly, establishing consensus across their individual taxonomies.

H1b: Consumers do not reach consensus in how they represent expenditures—people's taxonomies are substantially different.

Also, if a hierarchical representation is unnatural to people, their taxonomy might be different across time because there could be large variances in how they choose to build a taxonomy out of expenditures. Consequently, if people form similar hierarchical representations of expenditures over time, it would be suggestive of a hierarchical representation. Similar representations over time is also indicative of some stability in the way people represent relationships between expenditures, which would make it easier for people to recruit the taxonomic distance when making spending and savings decisions.

Second, a taxonomic representation predicts how much people will adjust their spending on items at different levels of the hierarchy. Specifically, we predict that, all else equal, consumers' spending adjustment on an item will be influenced by the taxonomic distance (i.e., the level at which items are categorized together) between the item they just spent on and an item they are considering purchasing. When a consumer deviates from her budget (i.e., overspends or underspends), she would adjust spending for items that are more closely related than those distantly related. If a consumer categorizes movies and scarves together as discretionary spending and treats bread as a necessity, the taxonomic distance between scarves and movies would be closer than that between scarves and bread. Consequently, we would predict that the consumer adjusts more on movies (i.e., the closer item) than on bread (i.e., the farther item).²

Several restrictions apply to the proposition that hierarchical taxonomy can predict spending adjustment. First, we do not suggest or believe that taxonomic distance is the only factor that consumers consider. Characteristics of expenditures such as the necessity of the spending will influence consumers spending intentions, and there are necessities that people cannot adjust onto such as rent or phone bills (Zhang et al. 2020). We argue that controlling for the other plausible factors, taxonomic distance will influence spending behaviors on items that have room for adjustment. Further, spending deviations from necessities might not allow room for choices on other items that need to be cut back on: if someone struggles to pay rent, then they will cut back on all the expenditures they possibly can. Therefore, we follow the previous literature (Heath and Soll 1996, Cheema and Soman 2006) and focus on spending adjustments in discretionary spending—that is, purchases that are not absolute necessities. Finally, we note that

² We note that this hypothesis holds regardless of whether H1a or H1b obtains. If H1b obtains, we can use information about each person's taxonomy of expenditures to predict her spending and saving decisions. If H1a obtains, we can use the consensus taxonomy to make predictions about a group's decisions.

we are attempting to characterize how people adjust their spending and savings when they are thinking about alternatives. This is the standard shopping context as people are almost always evaluating alternatives on product shelves or recommended lists on webpages. Therefore, we will always present alternatives for people to evaluate in our studies.

H2: All else equal, when consumers deviate from their budget, their taxonomy of expenditures influences their spending and saving decisions. Specifically, when they deviate from their budget (i.e., over- or under-spend) on an item, they are likely to adjust more on items that are closer in taxonomic distance to the purchased item.

Finally, if people naturally represent expenditures in hierarchically-organized categories, they will spontaneously adjust their spending in accordance with taxonomic distances. Here we use the term “spontaneous” to highlight that—contrary to previous literature (Heath and Soll 1996, Cheema and Soman 2006)—we do not prompt participants with predefined categories. We predict that consumers will make spending adjustments even when not explicitly prompted with budget category labels.

H3: People spontaneously recruit their taxonomy when making spending decisions. That is, the adjustment stated in H2 happens even without reminding people of the categories to which the expenditures belong.

In what follows, we first investigate how expenditures are represented. Then, we assess whether people spontaneously adjust their spending based on their representations even when they are not primed to think about budgeting categories.

Study Overview

In seven studies, we examine how consumers represent expenditures and how their representation of expenditures influences their budgetary decisions. In study 1a, we elicit

consumers' represented taxonomy of expenditures and find consensus in their representations. We find that people's representations are relatively stable across time in Study 1b. Study 2a and 2b investigate how people make spending and savings adjustments based on both each respondent's own individual taxonomy and the aggregate taxonomy, and Study 2c finds that people's hierarchical taxonomies influence their budgeting behaviors in a way not accounted for by the principles of substitutability and complementarity. The results are consistent with the hypothesis that people adjust their spending more on items that are taxonomically closer (whether the items are substitutes, complements, both, or neither). Studies 3a and 3b use consequential choices to test whether similar patterns generalize to purchasing items on promotion. Finally, Study 4 examines more than seven million trips to grocery stores across a decade, and finds analogous patterns—consumers spend more on items when taxonomically close items are on sale. The findings from all studies suggest that mental representation is important for studying how people budget and spend. The Web Appendix and all lab study materials (pre-registration, data, and analyses) are available on OSF (https://osf.io/bcknx/?view_only=94e720c38d014813bd1f672682d5b9a2).

Study 1: Representation of Expenditures

Study 1a investigates whether there is consensus in people's representations (H1) of expenditures while study 1b investigates the consistency of the representations across 3 months. Regardless of whether consensus is obtained, we will further qualitatively examine how consumers represent expenditures.

Study 1a: Initial Exploration

Participants & Procedures. Twenty-seven participants (all located in the US) were recruited on Amazon Mechanical Turk (MTurk). The average age was 38.4 and 37.0% were